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CHARMS checklist

CHARMS checklist. Relevant items to extract from individual studies for purposes of description and assessment of risk of bias and applicability. (Adapted from Moons KG. 2014)

Gen	App	RoB	Domain/ Key items		Gaca, 2011		De Feo, 2012		Martínez-Sellés, 2014		Madeira, 2016		Gatti (a), 2017		Gatti (
0. Study information															
			0.1 Author		Gaca		De Feo		Martínez-Sellés		Madeira		Gatti (a)		Ga
			0.2 Publication year		2011		2012		2014		2016		2017		2
			0.3 Publication identifier (Title/PMID/DOI)		Outcomes for endocarditis surgery in North		The Need for a Specific Risk Prediction System in		Valve surgery in active infective endocarditis: A		Assessment of perioperative mortality risk in		Simple Scoring System to Predict In-Hospital		Simple Scoring Syster
			0.4 Publication journal		J Thorac Cardiovasc Surg		The Scientific World Journal		International Journal of Cardiology		Interactive CardioVascular and Thoracic Surgery		Journal of American Heart Association		Journal of Americ
			0.5 Model name		STSS score		De Feo score		PALSUSE		modified EuroSCORE-II		AEPEI score		Alternate
1. Source of data					1		1		1		1		1		1
			1.1 Source of data		Existing registry		Retrospective cohort		Existing registry		Retrospective cohort		Other (specify)		Other
2. Participants					1		1		1		1		1		1
			2.1 Recruitment method		Selective inclusion		Prospective recruitment		Consecutive patients		No information		Consecutive patients		Consecut
			2.2 Recruitment dates		2002 - 2008		1980 - 2009		2008 - 2010		2007 - 2014		2000 - 2015 (Italy) 2008 (France)		2000 - 2015 (Ita
			2.3 Study setting		Cardiac surgery centers		Cardiac surgery center		Cardiac surgery centers		Cardiac surgery center		Cardiac surgery centers		Cardiac su
			2.4 Study sites (Regions)		North America		Italy		Spain		Portugal		Italy and France		Italy ar
			2.5 Study sites (Number of centers)		Unclear		1		26		1		8		
			2.6 Criteria inclusion		All patients with the diagnosis of IE who underwent surgery on the aortic, mitral, and/or tricuspid valves.		Infective Endocarditis of native valves		Patients with definite or possible IE who underwent surgery.		Patients underwent cardiac surgery under extracorporeal circulation for active IE (according to modified Duke criteria).		Patients who underwent surgery for IE defined by the modified Duke criteria		Patients who underwent the modifie
			2.7 Criteria exclusion		Sites were excluded if data were missing on age, gender, status of surgery, cardiogenic shock , and endocarditis type. And if more than 20% of patients had no complication information reported.		Infective Endocarditis of prosthetic valves		No information		No information		No information		No info
			2.8 Participant description	*Characteristics*	*Values*	*Measures*	*Values*	*Measures*	*Values*	*Measures*	*Values*	*Measures*	*Values*	*Measures*	*Values*
			2.8.1 Age of participants	*Age of participants*	55 (46;66)	Median (IQR)	49 (16)	Mean (sd)	61.4 (15.5)	Mean (sd)	60 (47;70)	Median (IQR)	59.1 (15.4)	Mean (sd)	59.1 (15.4)
			2.8.2 Native valve endocarditis	*Native valve endocarditis*	No information	No information	440 (100)	n (%)	267 (61.1)	n (%)	94 (73.4)	n (%)	285 (78.9)	n (%)	285 (78.9)
			2.8.3 Valve affected	*Valve affected*	All	Other	All	Other	All	Other	All	Other	All	Other	All
			2.8.4 Characteristic 4												
			2.8.5 Characteristic 5												
3. Outcome to be predicted					1		1		1		1		1		1
			3.1 Outcome		In-hospital or 30 days mortality		In-hospital mortality or 30 days mortality		In-hospital mortality		In-hospital or 30 days mortality		In-hospital mortality		In-hospit
			3.2 Outcome definition		Death occurring before discharge or within 30 days of surgery.		30-day death or in hospital death		Operative mortality ocurring during the admission		at 30 days or during the index hospitalization		In-hospital death		In-hosp
			3.3 Same outcome definition for all participants		Yes		Yes		Yes		Yes		Yes		Y
			3.4 Type of outcome		Single		Single		Single		Single		Single		Si
			3.5 Was the outcome assessed without knowledge of the predictors?		Unclear		Unclear		Unclear		Unclear		Unclear		Un
			3.6 Were candidate predictors part of outcome?		No		No		No		No information		No		I
			3.7 Time of outcome ocurrence		30 days or lenght of hospital stay		30 days or lenght of hospital stay		Lenght of hospital stay		30 days or lenght of hospital stay		Lenght of hospital stay		Lenght of
4. Candidate predictors					1		1		1		1		1		1
			4.1 Number of candidate predictors (or parameters) assessed		38		19				15		57		
			4.2 Type of predictors		Patient, surgery and IE related factors		Patient, surgery and IE related factors		Patient, surgery and IE related factors		Patient and IE related factors		Patient related factors		Patient rel
			4.3 Timing of predictors measurement		Pre-operative		pre-operative		Pre-operative		Pre-operative		Pre-operative		Pre-o
			4.4 Predictors definition and measurement similar for all participants		Yes		Yes		Yes		Yes		Unclear		Un
			4.5 Were predictors assessed blinded for outcome?		No information		No information		No information		No information		No information		No info
			4.6 Handling of continuous predictors		No information		No information		Dichotomization		No information		No information		No info
5. Sample size					1		1		1		1		1		1
			5.1 Number of participants		13,617		440		437		128		361		3
			5.2 Number of outcomes/events		1,117		40		106		21		56		
			5.3 Number events per variable (EPV) or per parameter (EPP)		29.4		2.1		Unknown		1.4		1.0		1
6. Missing data					1		1		1		1		1		1
			6.1 Number of participants with any missing value		98		22		No information		No information		No information		No info
			6.2 Handling of missing data		No information		No information		No information		No information		No information		No info
7. Model development					1		1		1		1		1		1
			7.1 Modelling method		Logistic GEE regression		Logistic regression		Logistic regression		Logistic regression		Logistic regression		Logistic
			7.2 Method for selection of candidate predictors		Based on univariable associations		Based on univariable associations		Based on univariable associations		Based on univariable associations		Based on univariable associations		Based on univar
			7.3 Method for selection of predictors during multivariable modelling		No information		No information		Stepwise selection		No information		Backward elimination		Backward
			7.4 Shrinkage of predictor weights or regression coefficients		No		No		No		No		No		I

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Gen	App	RoB	Domain/ Key items	Gaca, 2011			De Feo, 2012			Martínez-Sellés, 2014			Madeira, 2016			Gatti (a), 2017			Gatti (		
8. Model performance				1			1			1			1			1			1		
			8.1 Calibration measures	Calibration plot			HL test			HL test			Calibration plot / Slope / CITL / HL test			HL test			HL		
			8.1.1 Calibration plot	Yes			No			No			Yes			No					
			8.1.2 Calibration slope	No	value (95% CI):		No	value (95% CI):		No	value (95% CI):		Yes	value (95% CI):	0.8	No	value (95% CI):		No	value (95% CI):	
			8.1.3 Calibration-in-the-large (CITL)	No	value (95% CI):		No	value (95% CI):		No	value (95% CI):		Yes	value (95% CI):	1.9	No	value (95% CI):		No	value (95% CI):	
			8.1.4 Hosmer-Lemeshow test	No			Yes			Yes			Yes			Yes			Y		
			8.1.5 Other	No	Specify:		No	Specify:		No	Specify:		No	Specify:		No	Specify:		No	Specify:	
			8.2 Discrimination measures	C-Statistic			C-Statistic / AUC graph			C-Statistic / AUC graph			C-Statistic / AUC graph			C-Statistic / AUC graph			C-Statistic		
			8.2.1 C-Statistic	Yes	value (95% CI):	0.758	Yes	value (95% CI):	0.88 (0.82 to 0.93)	Yes	value (95% CI):	0.84 (0.79 to 0.88)	Yes	value (95% CI):	0.87 (0.79 to 0.94)	Yes	value (95% CI):	0.72 (0.64 to 0.78)	Yes	value (95% CI):	
			8.2.2 D-Statistic	No	value (95% CI):		No	value (95% CI):		No	value (95% CI):		No	value (95% CI):		No	value (95% CI):		No	value (95% CI):	
			8.2.3 AUC graph	No			Yes			Yes			Yes			Yes			Y		
			8.2.4 Log-rank test (if survival analysis)	Not applicable			Not applicable			Not applicable			Not applicable			Not applicable			Not ap		
			8.2.5 Risk group curves (if survival analysis)	Not applicable			Not applicable			Not applicable			Not applicable			Not applicable			Not ap		
			8.2.6 Other	No	Specify:		No	Specify:		No	Specify:		No	Specify:		No	Specify:		No	Specify:	
			8.3 Overall measures	Not evaluated			Not evaluated			Not evaluated			Brier score			Not evaluated			Not ev		
			8.3.1 R-squared (e.g. Cox-Snell R2, Nagelkerke's R2,..)	No	value (95% CI):		No	value (95% CI):		No	value (95% CI):		No	value (95% CI):		No	value (95% CI):		No	value (95% CI):	
			8.3.2 Brier score	No	value (95% CI):		No	value (95% CI):		No	value (95% CI):		Yes	value (95% CI):	0.1	No	value (95% CI):		No	value (95% CI):	
			8.3.3 Other	No	Specify:		No	Specify:		No	Specify:		No	Specify:		No	Specify:		No	Specify:	
			8.4 Clinical utility	Not evaluated			Not evaluated			Not evaluated			Not evaluated			Not evaluated			Not ev		
			8.4.1 Decision Curve Analysis (DCA)	No			No			No			No			No					
			8.4.2 Other	No	Specify:		No	Specify:		No	Specify:		No	Specify:		No	Specify:		No	Specify:	
9. Model evaluation				1			1			1			1			1			1		
			9.1 Method used for testing model performance																		
			9.1.1 Internal validation	Random split data			None (Apparent performance)			None (Apparent performance)			None (Apparent performance)			Bootstrap			Boo		
			9.1.2 External validation	None			None			None			None			Geographical			N		
			9.2 In case of poor validation, whether model was adjusted or updated	No			No			No			No			No					
10. Results				1			1			1			1			1			1		
			10.1 Number of predictors (or parameters) included in final model	13			6			7			2			5					
			10.2 Final model included predictor weights or regression coefficients	Regression coefficients			Regression coefficients			Regression coefficients			Regression coefficients			Regression coefficients			Regressior		
			10.3 Final model included intercept (or baseline survival)	No			No			No			No			Yes					
			10.4 Alternative presentation of the final prediction models	Score system			Score system			Score system			None			Score system			Score		
11. Interpretation				1			1			1			1			1			1		
			11.1 Interpretation of presented model	The risk scoring system informs patient selection, provides risk stratification, and allows communication with patients and other physicians.			Our score may serve to this scope, provided that it will be validated in other larger series, possibly in multicentric design.			The PALSUSE score could help to identify patients with higher in-hospital mortality following IE surgery.			-			Further larger validation studies are necessary before introducing the AEPEI score into the clinical practice.			Further larger validatic before introducing the A pra		
12. Observations				1			1			1			1			1			1		
			12.1 Data extraction process	All information has been successfully registered			All information has been successfully registered			All information has been successfully registered			All information has been successfully registered			All information has been successfully registered			All information has be		
			12.2 Additional information	Modelling method was Logistic mixed effect model												Source of was a combination beetwen a retrospective cohort from Italy and a existing registry from France.			Source of was a combinat retrospective cohort from registry from France.		

Abbreviation: *Gen: General description; App: Applicability; RoB: Risk of Bias*

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Gen	App	RoB	Domain/ Key items		(b), 2017	Di Mauro, 2017		Gatti (c), 2017		Olmos, 2017		Fernández-Hidalgo (a), 2018		Fernández-Hidalgo (b), 2018	
0. Study information															
			0.1 Author		tti (b)	Di Mauro		Gatti (c)		Olmos		Fernández-Hidalgo (a)		Fernández-Hidalgo (b)	
			0.2 Publication year		017	2017		2017		2017		2018		2018	
			0.3 Publication identifier (Title/PMID/DOI)		n to Predict In-Hospital	A predictive model for early mortality after surgical		A risk factor analysis for in-hospital mortality after		Risk score for cardiac surgery in active left-sided		A pragmatic approach for mortality prediction after		A pragmatic approach for mortality prediction after	
			0.4 Publication journal		an Heart Association	Int J Cardiol		Infection		Heart		Clinical Microbiology and Infection		Clinical Microbiology and Infection	
			0.5 Model name		AEPEI score	EndoSCORE		ANCLA score		RISK-E score		Specific EuroSCORE-I		Specific EuroSCORE-II	
1. Source of data						1		1		1		1		1	
			1.1 Source of data		(specify)	Retrospective cohort		Retrospective cohort		Retrospective cohort		Retrospective cohort		Retrospective cohort	
2. Participants						1		1		1		1		1	
			2.1 Recruitment method		ive patients	No information		Consecutive patients		Consecutive patients prospectively recruited on an		Consecutive patients		Consecutive patients	
			2.2 Recruitment dates		aly) 2008 (France)	2000 - 2015		1999 - 2015		1996 - 2014		2000 - 2011		2000 - 2011	
			2.3 Study setting		rgery centers	Cardiac surgery centers		Cardiac surgery center		Cardiac surgery centers		Cardiac surgery centers		Cardiac surgery centers	
			2.4 Study sites (Regions)		nd France	Italy		Italy		Spain		Spain		Spain	
			2.5 Study sites (Number of centers)		8	26		1		3		9		9	
			2.6 Criteria inclusion		t surgery for IE defined by d Duke criteria	Patients with native valve endocarditis (NVE) or prosthesis valve endocarditis (PVE) operated.		Patients who underwent surgery for IE defined by the modified Duke criteria.		Patients with definite, left-sided IE in the active phase of the disease who underwent surgery.		Patients who requiring cardiac surgery during the active phase of infection (while on antimicrobial therapy).		Patients who requiring cardiac surgery during the active phase of infection (while on antimicrobial therapy).	
			2.7 Criteria exclusion		ormation	No information		No information		Patients with right-sided IE, left-sided IE without surgical indications and those with a surgical indication that were not surgical candidates.		IE cases exclusively involving pacemaker leads were not considered. Only the first episode of IE recorded for an individual patient was used.		IE cases exclusively involving pacemaker leads were not considered. Only the first episode of IE recorded for an individual patient was used.	
			2.8 Participant description	*Characteristics*	*Measures*	*Values*	*Measures*	*Values*	*Measures*	*Values*	*Measures*	*Values*	*Measures*	*Values*	*Measures*
			2.8.1 Age of participants	*Age of participants*	Mean (sd)	59.6 (15.1)	Mean (sd)	60.6 (8.5)	Mean (sd)	62 (14)	Mean (sd)	58 (15.1)	Mean (sd)	58 (15.1)	Mean (sd)
			2.8.2 Native valve endocarditis	*Native valve endocarditis*	n (%)	2.221 (82)	n (%)	103 (74.6)	n (%)	259 (61.1)	n (%)	No information	No information	No information	No information
			2.8.3 Valve affected	*Valve affected*	Other	All	Other	All	Other	Aortic / Mitral	Other	All	Other	All	Other
			2.8.4 Characteristic 4												
			2.8.5 Characteristic 5												
3. Outcome to be predicted						1		1		1		1		1	
			3.1 Outcome		al mortality	30 days mortality		In-hospital mortality		In-hospital mortality		30 days mortality		30 days mortality	
			3.2 Outcome definition		ital death	Death by 30 days after surgery due to any cause		In-hospital postsurgery mortality		Death ocurring after surgery and before discharge, regardless of its cause.		Death occurring within the first 30 days after cardiac surgery.		Death occurring within the first 30 days after cardiac surgery.	
			3.3 Same outcome definition for all participants		Yes	Yes		Yes		Yes		Yes		Yes	
			3.4 Type of outcome		ngle	Single		Single		Single		Single		Single	
			3.5 Was the outcome assessed without knowledge of the predictors?		clear	Unclear		Unclear		Unclear		Unclear		Unclear	
			3.6 Were candidate predictors part of outcome?		No	No information		No		No		No		No	
			3.7 Time of outcome ocurrence		hospital stay	30 days		Lenght of hospital stay		Lenght of hospital stay		30 days		30 days	
4. Candidate predictors						1		1		1		1		1	
			4.1 Number of candidate predictors (or parameters) assessed		57	32		56		37		26		27	
			4.2 Type of predictors		lated factors	Patient, surgery and IE related factors		Patients and surgery related factors		Patient, surgery and IE related factors		Patient, surgery and IE related factors		Patient, surgery and IE related factors	
			4.3 Timing of predictors measurement		perative	Pre-operative		Pre-operative		Pre-operative		Pre-operative		Pre-operative	
			4.4 Predictors definition and measurement similar for all participants		clear	Yes		Yes		Yes		No		No	
			4.5 Were predictors assessed blinded for outcome?		ormation	No information		No information		No information		No information		No information	
			4.6 Handling of continuous predictors		ormation	Restricted cubic spline function or categorization		No information		No information		No information		No information	
5. Sample size						1		1		1		1		1	
			5.1 Number of participants		861	2,715		138		424		779		779	
			5.2 Number of outcomes/events		56	298		28		124		208		208	
			5.3 Number events per variable (EPV) or per parameter (EPP)		1.0	9.3		0.5		3.4		8.0		7.7	
6. Missing data						1		1		1		1		1	
			6.1 Number of participants with any missing value		ormation	No information		45		No information		4		4	
			6.2 Handling of missing data		ormation	No information		No information		No information		Complete-case analysis		Complete-case analysis	
7. Model development						1		1		1		1		1	
			7.1 Modelling method		regression	Logistic regression		Logistic regression		Logistic regression		Logistic regression		Logistic regression	
			7.2 Method for selection of candidate predictors		riable associations	Based on univariable associations		Based on univariable associations		Based on univariable associations and clinical		Based on prior knowledge		Based on prior knowledge	
			7.3 Method for selection of predictors during multivariable modelling		elimination	No information		Backward elimination		Stepwise selection		Bootstrap selection		Bootstrap selection	
			7.4 Shrinkage of predictor weights or regression coefficients		No	No		No		No		No		No	

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Gen	App	RoB	Domain/ Key items	b), 2017	Di Mauro, 2017		Gatti (c), 2017		Olmos, 2017		Fernández-Hidalgo (a), 2018		Fernández-Hidalgo (b), 2018	
8. Model performance					1		1		1		1		1	
			8.1 Calibration measures	test	Comparison of actual CITL and slope with the ideal		HL test		Calibration plot / HL test		Calibration plot / Slope / CITL		Calibration plot / Slope / CITL	
			8.1.1 Calibration plot	No	No		No		Yes		Yes		Yes	
			8.1.2 Calibration slope		No	value (95% CI):	No	value (95% CI):	No	value (95% CI):	Yes	value (95% CI): 0.93 (0.92 to 0.93)	Yes	value (95% CI): 0.93 (0.92 to 0.94)
			8.1.3 Calibration-in-the-large (CITL)		No	value (95% CI):	No	value (95% CI):	No	value (95% CI):	Yes	value (95% CI): -0.06 (-0.07 to -0.05)	Yes	value (95% CI): -0.05 (-0.06 to -0.04)
			8.1.4 Hosmer-Lemeshow test	Yes	No		Yes		Yes		No		No	
			8.1.5 Other		Yes	Specify: Comparison of actual CITL	No	Specify:	No	Specify:	No	Specify:	No	Specify:
			8.2 Discrimination measures	/ AUC graph	C-Statistic / AUC graph		C-Statistic / AUC graph		C-Statistic / AUC graph		C-Statistic / AUC graph		C-Statistic / AUC graph	
			8.2.1 C-Statistic	0.69 (0.61 to 0.76)	Yes	value (95% CI): 0.85 (0.84 to 0.86)	Yes	value (95% CI): 0.83 (0.75 to 0.89)	Yes	value (95% CI): 0.76 (0.64 to 0.88)	Yes	value (95% CI): 0.77 (0.74 to 0.81)	Yes	value (95% CI): 0.77 (0.73 to 0.81)
			8.2.2 D-Statistic		No	value (95% CI):	No	value (95% CI):	No	value (95% CI):	No	value (95% CI):	No	value (95% CI):
			8.2.3 AUC graph	Yes	Yes		Yes		Yes		Yes		Yes	
			8.2.4 Log-rank test (if survival analysis)	pplicable	Not applicable		Not applicable		Not applicable		Not applicable		Not applicable	
			8.2.5 Risk group curves (if survival analysis)	pplicable	Not applicable		Not applicable		Not applicable		Not applicable		Not applicable	
			8.2.6 Other		No	Specify:	No	Specify:	No	Specify:	No	Specify:	No	Specify:
			8.3 Overall measures	valuated	Brier score		Not evaluated		Not evaluated		Brier score		Brier score	
			8.3.1 R-squared (e.g. Cox-Snell R2, Nagelkerke's R2,..)		No	value (95% CI):	No	value (95% CI):	No	value (95% CI):	No	value (95% CI):	No	value (95% CI):
			8.3.2 Brier score		Yes	value (95% CI): 0.065 (0.057 to 0.072)	No	value (95% CI):	No	value (95% CI):	Yes	value (95% CI): 0.159	Yes	value (95% CI): 0.159
			8.3.3 Other		No	Specify:	No	Specify:	No	Specify:	No	Specify:	No	Specify:
			8.4 Clinical utility	valuated	Not evaluated		Not evaluated		Not evaluated		Not evaluated		Not evaluated	
			8.4.1 Decision Curve Analysis (DCA)	No	No		No		No		No		No	
			8.4.2 Other		No	Specify:	No	Specify:	No	Specify:	No	Specify:	No	Specify:
9. Model evaluation					1		1		1		1		1	
			9.1 Method used for testing model performance											
			9.1.1 Internal validation	tstrap	Bootstrap		Bootstrap		Random split data		Bootstrap		Bootstrap	
			9.1.2 External validation	one	None		None		Geographical		None		None	
			9.2 In case of poor validation, whether model was adjusted or updated	No	No		No		No		No		No	
10. Results					1		1		1		1		1	
			10.1 Number of predictors (or parameters) included in final model	3	15		5		8		10		9	
			10.2 Final model included predictor weights or regression coefficients	coefficients	Regression coefficients		Regression coefficients		Regression coefficients		Predictor weights		Predictor weights	
			10.3 Final model included intercept (or baseline survival)	No	Yes		No		Yes		No		No	
			10.4 Alternative presentation of the final prediction models	system	None		Score system		Score system		Nomogram		Nomogram	
11. Interpretation					1		1		1		1		1	
			11.1 Interpretation of presented model	on studies are necessary EPEI score into the clinical ctice.	GIROC provides a logistic risk model to predict early mortality in patients with heart valve or prosthesis IE undergoing surgery.		ANCLA score could aid a rapid and reliable framing of the patient with IE. Further large validation studies are necessary.		This risk-scoring model should be a useful tool for clinical decision-making, providing more accurate information to patients and their families.		We suggest that a new risk score based on EuroSCORE I should be used for prospective validation.		We suggest that a new risk score based on EuroSCORE I should be used for prospective validation.	
12. Observations					1		1		1		1		1	
			12.1 Data extraction process	en successfully registered	All information has been successfully registered		All information has been successfully registered		All information has been successfully registered		All information has been successfully registered		All information has been successfully registered	All i
			12.2 Additional information	tion beetwen a Italy and a existing										

Abbreviation: *Gen: General description; App: Applicability; RoB: Risk of Bias*

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Gen	App	RoB	Domain/ Key items																
0. Study information																			
			0.1 Author																
			0.2 Publication year																
			0.3 Publication identifier (Title/PMID/DOI)																
			0.4 Publication journal																
			0.5 Model name																
1. Source of data																			
			1.1 Source of data																
2. Participants																			
			2.1 Recruitment method																
			2.2 Recruitment dates																
			2.3 Study setting																
			2.4 Study sites (Regions)																
			2.5 Study sites (Number of centers)																
			2.6 Criteria inclusion																
			2.7 Criteria exclusion																
			2.8 Participant description	*Characteristics*	*Values*	*Measures*	*Values*	*Measures*	*Values*	*Measures*	*Values*	*Measures*	*Values*	*Measures*	*Values*
			2.8.1 Age of participants	*Age of participants*											
			2.8.2 Native valve endocarditis	*Native valve endocarditis*											
			2.8.3 Valve affected	*Valve affected*											
			2.8.4 Characteristic 4												
			2.8.5 Characteristic 5												
3. Outcome to be predicted																			
			3.1 Outcome																
			3.2 Outcome definition																
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			3.4 Type of outcome																
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			4.1 Number of candidate predictors (or parameters) assessed																
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			4.5 Were predictors assessed blinded for outcome?																
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6. Missing data																			
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			6.2 Handling of missing data																
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			7.1 Modelling method																
			7.2 Method for selection of candidate predictors																
			7.3 Method for selection of predictors during multivariable modelling																
			7.4 Shrinkage of predictor weights or regression coefficients																

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8. Model performance									
			8.1 Calibration measures						
			8.1.1 Calibration plot						
			8.1.2 Calibration slope	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.1.3 Calibration-in-the-large (CITL)	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.1.4 Hosmer-Lemeshow test						
			8.1.5 Other	Specify:	Specify:	Specify:	Specify:	Specify:	Specify:
			8.2 Discrimination measures						
			8.2.1 C-Statistic	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.2.2 D-Statistic	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.2.3 AUC graph						
			8.2.4 Log-rank test (if survival analysis)						
			8.2.5 Risk group curves (if survival analysis)						
			8.2.6 Other	Specify:	Specify:	Specify:	Specify:	Specify:	Specify:
			8.3 Overall measures						
			8.3.1 R-squared (e.g. Cox-Snell R2, Nagelkerke's R2,..)	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.3.2 Brier score	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.3.3 Other	Specify:	Specify:	Specify:	Specify:	Specify:	Specify:
			8.4 Clinical utility						
			8.4.1 Decision Curve Analysis (DCA)						
			8.4.2 Other	Specify:	Specify:	Specify:	Specify:	Specify:	Specify:
9. Model evaluation									
			9.1 Method used for testing model performance						
			9.1.1 Internal validation						
			9.1.2 External validation						
			9.2 In case of poor validation, whether model was adjusted or updated						
10. Results									
			10.1 Number of predictors (or parameters) included in final model						
			10.2 Final model included predictor weights or regression coefficients						
			10.3 Final model included intercept (or baseline survival)						
			10.4 Alternative presentation of the final prediction models						
11. Interpretation									
			11.1 Interpretation of presented model						
12. Observations									
			12.1 Data extraction process	nformation has been successfully registered	All information has been successfully registered	All information has been successfully registered	All information has been successfully registered	All information has been successfully registered	All information has be
			12.2 Additional information						

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Gen	App	RoB	Domain/ Key items												
0. Study information															
			0.1 Author												
			0.2 Publication year												
			0.3 Publication identifier (Title/PMID/DOI)												
			0.4 Publication journal												
			0.5 Model name												
1. Source of data															
			1.1 Source of data												
2. Participants															
			2.1 Recruitment method												
			2.2 Recruitment dates												
			2.3 Study setting												
			2.4 Study sites (Regions)												
			2.5 Study sites (Number of centers)												
			2.6 Criteria inclusion												
			2.7 Criteria exclusion												
			2.8 Participant description	*Characteristics*	*Measures*	*Values*	*Measures*	*Values*	*Measures*	*Values*	*Measures*	*Values*	*Measures*	*Values*	*Measures*
			2.8.1 Age of participants	*Age of participants*											
			2.8.2 Native valve endocarditis	*Native valve endocarditis*											
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			8.1.1 Calibration plot						
			8.1.2 Calibration slope		value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.1.3 Calibration-in-the-large (CITL)		value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.1.4 Hosmer-Lemeshow test						
			8.1.5 Other		Specify:	Specify:	Specify:	Specify:	Specify:
			8.2 Discrimination measures						
			8.2.1 C-Statistic		value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.2.2 D-Statistic		value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.2.3 AUC graph						
			8.2.4 Log-rank test (if survival analysis)						
			8.2.5 Risk group curves (if survival analysis)						
			8.2.6 Other		Specify:	Specify:	Specify:	Specify:	Specify:
			8.3 Overall measures						
			8.3.1 R-squared (e.g. Cox-Snell R2, Nagelkerke's R2,..)		value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.3.2 Brier score		value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.3.3 Other		Specify:	Specify:	Specify:	Specify:	Specify:
			8.4 Clinical utility						
			8.4.1 Decision Curve Analysis (DCA)						
			8.4.2 Other		Specify:	Specify:	Specify:	Specify:	Specify:
9. Model evaluation									
			9.1 Method used for testing model performance						
			9.1.1 Internal validation						
			9.1.2 External validation						
			9.2 In case of poor validation, whether model was adjusted or updated						
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			11.1 Interpretation of presented model						
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			2.8.1 Age of participants	*Age of participants*											
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			8.1.3 Calibration-in-the-large (CITL)	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
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			8.1.5 Other	Specify:	Specify:	Specify:	Specify:	Specify:	Specify:
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			8.2.1 C-Statistic	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.2.2 D-Statistic	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):	value (95% CI):
			8.2.3 AUC graph						
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			8.2 Discrimination measures				
			8.2.1 C-Statistic		value (95% CI):		value (95% CI):
			8.2.2 D-Statistic		value (95% CI):		value (95% CI):
			8.2.3 AUC graph				
			8.2.4 Log-rank test (if survival analysis)				
			8.2.5 Risk group curves (if survival analysis)				
			8.2.6 Other		Specify:		Specify:
			8.3 Overall measures				
			8.3.1 R-squared (e.g. Cox-Snell R2, Nagelkerke's R2,..)		value (95% CI):		value (95% CI):
			8.3.2 Brier score		value (95% CI):		value (95% CI):
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